Bureau
International des
Poids et
Mesures

Le système international d'unités
The International System of Units

9 édition 2019

Annexe 2
Appendix 2

Valeurs recommandées des fréquences étalons destinées à la mise en pratique de la définition du mètre et aux représentations secondaires de la seconde

Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second

Partie 2.6
Part 2.6

Ytterbium 171 Ion ($f \approx 688\,\mathrm{THz}$)

Ytterbium 171 Ion ($f \approx 688\,\mathrm{THz}$)

May 2016
mai 2016

Bureau International
des Poids et Mesures

Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second Part 2.6: Ytterbium 171 Ion ($f \approx 688\ \mathrm{THz}$)

9th edition 2019

Appendix 2 May 2016

$^{171}Yb^{+}$ ion, $6s\,^2S_{1/2} - 5d\,^2D_{3/2}$ unperturbed optical transition

1. Recommended value [1] of the frequency

$f(\mathrm{Yb}^+, \text{quadrupole}) = 688\,358\,979\,309\,308.3\ \mathrm{Hz}$

equivalent to

$\lambda(\mathrm{Yb}^+) = 435\,517\,610.739\,688\ \mathrm{fm}$

with an estimated relative standard uncertainty of 6×10^{-16}.

This radiation is already endorsed by the CIPM as a secondary representation of the definition of the second.

2. Source data

Adopted value $f(\mathrm{Yb}^+) = 688\,358\,979\,309\,308.3$ Hz

$u_c/y = 6 \times 10^{-16}$

calculated from

f (^{171}Yb$^+$, quadrupole) / Hz	u / Hz	f(^{171}Yb$^+$ oct)/ f(^{171}Yb$^+$ quad)	u	source data
688 358 979 309 308	2.14			[2]
688 358 979 309 306.97	0.73			[3]
688 358 979 309 310	9			[4]
688 358 979 309 308.42	0.42			[5]
688 358 979 309 307.82	0.36			[6]
		0.932 829 404 530 964 65	3.1×10^{-16}	[5]

as a weighted mean from the above values applying the method of Margolis and Gill [7]. Considering that the values came from only two laboratories, the CCTF considered it prudent to enlarge the uncertainty by a factor of two.

References

[1] CIPM Recommendation 2 (CI-2015): Updates to the list of standard frequencies http://www.bipm.org/jsp/en/CIPMRecommendations.jsp

[2] Report of the 12th meeting, Consultative Committee for Length (CCL), 15-16 Sept. 2005; available at http://www.bipm.org/utils/common/pdf/CCL12.pdf and Questionnaire previous to the 2006 meeting of the CCL/CCTF Joint Working GroupCCL-CCTF/06-11.

[3] Chr. Tamm, S. Weyers, B. Lipphardt, and E. Peik: Stray-field-induced quadrupole shift and absolute frequency of the 688-THz $^{171}Yb^{+}$ single-ion optical frequency standard, *Phys. Rev. A* **80**, 043403 (2009).

[4] S. Webster, R. Godun, S. King, G. Huang, B. Walton, V. Tsatourian, H. Margolis, S. Lea and P. Gill: Frequency Measurement of the 2S1/2–2D3/2 Electric Quadrupole Transition in a Single $^{171}Yb^{+}$ Ion, *IEEE Trans.on Ultrasonics, Ferroelectrics, and Frequency Control*, **57**, 592 (2010).

[5] Godun, R. M.; Nisbet-Jones, P. B. R.; Jones, J. M.; King, S. A.; Johnson, L. A. M.; Margolis, H. S.; Szymaniec, K.; Lea, S. N.; Bongs, K. & Gill, P.: Frequency ratio of two optical clock transitions in $^{171}Yb^{+}$ and constraints on the time-variation of fundamental constants, *Phys. Rev. Lett.*, **113**, 210801 (2014).

[6] Tamm, C.; Huntemann, N.; Lipphardt, B.; Gerginov, V.; Nemitz, N.; Kazda, M.; Weyers, S. & Peik, E.: A Cs-Based Optical Frequency Measurement Using Cross-Linked Optical and Microwave Oscillators, *Phys. Rev.A*, **89**, 023820 (2014).

[7] H. S. Margolis, P. Gill: Least-squares analysis of clock frequency comparison data to deduce optimized frequency and frequency ratio values; *Metrologia*, 2015, **52**, 628-634